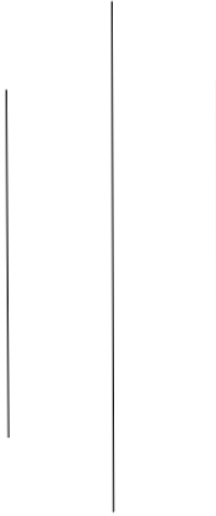


A technical Report on Fluid Mechanics



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Fluid mechanics

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Contents

Part-1 Bernoulli's Theorem Experiment	4
1.1 Objective	4
1.2 Apparatus Used	4
1.3. Theory	4
1.3.1 Head Measurement	5
1.3.2 Velocity Measurement.....	5
1.3.3 Test procedure.....	5
1.4 Results and Discussion	6
1.4.2. Table of Results	6
1.4.3 Graphs	8
1.5. conclusion and discussion of any discrepancies observed.....	9
1.5.1 Conclusion.....	9
1.5.2 Discrepancies Observed	9
Part 2- Hydraulic Modelling of Flow Over a Weir in an Open Channel.....	10
2.1. Objective	10
2.2. Apparatus used.....	10
2.3. Theory	10
2.3.1 Test Procedure.....	11
2.4 Result and Discussion	11
2.4.1. Table of Raw Data	11
2.4.2 Calculation	12
2.4.3 Site photos	12
2.4.4 Graph.....	12
2.4.5Comparison of Experimental Results with Theory	12
2.4.6 Comparison with Released Cd Values	13
2.4.7 Observation and sources of error	13
2.4.8 Dimensional and Similarity Considerations	14
2.5 Conclusion	14

Part 3. Centrifugal Pump Performance & Selection	15
3.1 Engineering Scenario.....	15
3.2 Pump Selection and criteria	15
3.2.1 Identifying which meets the required flow and head	15
3.2.2 Determining the operating efficiency and power requirement.....	16
3.2.3 Graphs	16
3.3 Discussion.....	17
3.3.1 Proximity to BEP.....	17
3.3.2 Risk of operating too far left/right of curve & Energy consumption implications.	17
3.3.3 Cavitation considerations.....	17
3.4 Conclusion and observation	17

Part-1 Bernoulli's Theorem Experiment

1.1 Objective

To show how pressure changes in a converging-diverging section and to confirm Bernoulli's theorem by experimentally proving that the total of the terms in the Bernoulli equation along a streamline stays constant.

1.2 Apparatus Used

1. F1-10 hydraulics bench
2. F1-15 Bernoulli's apparatus test equipment
3. A stopwatch for timing the flow measurement

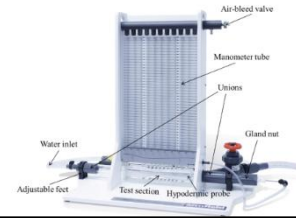


Fig: Bernoulli's Test Apparatus

1.3. Theory

Bernoulli's theorem states that for an ideal fluid that is incompressible, non-viscous, and flowing steadily, the total mechanical energy per unit weight remains constant along a streamline (Pritchard & Mitchell, 2016)

In other words, the sum of pressure energy, kinetic energy, and potential energy remains unchanged as the fluid moves through a flow system. This principle is fundamental in fluid mechanics and is widely used to analyze the behavior of fluids in pipes, channels, and hydraulic devices (Qin & Duan, 2017)

Physically, it implies that when the velocity of a fluid increases, its pressure decreases, and when the velocity decreases, the pressure increases, provided there are no energy losses. This relationship is clearly observed in devices such as Venturi meters, where a reduction in cross-sectional area leads to an increase in velocity and a corresponding drop in pressure (Wahba, 2022). The Bernoulli's equation is expressed as:

$$p + \frac{1}{2}\rho v^2 + \rho g z = p_0 = \text{constant}$$

$$\frac{p}{\rho g} + \frac{v^2}{2g} + z = h_t = \text{constant}$$

Where,

- p = pressure (Pa or N/m²)
- ρ = density of the fluid (kg/m³)
- v = velocity of flow (m/s)
- p_0 = total pressure (Pa or N/m²)

- g = acceleration due to gravity (9.81 m/s²)
- z = elevation from datum (m)
- h_t = total head (m)

1.3.1 Head Measurement

In fluid mechanics, head measurement is a way to show the energy of a fluid based on the height of a fluid column. This method helps in comparing various energy types using a standard unit (meters). The total head of a moving fluid consists of pressure head, velocity head, and elevation head. According to Bernoulli's principle, this total stays the same along a streamline when conditions are steady, incompressible, and frictionless (Pritchard & Mitchell, 2016). The total head is given by:

$$h_t = h + \frac{V^2}{2g} \quad \text{Where,}$$

- ✚ h_t = total head (m)
- ✚ h = pressure head (m)
- ✚ V = velocity of flow (m/s)
- ✚ g = acceleration due to gravity (9.81 m/s²)

1.3.2 Velocity Measurement

Flow velocity is determined from the volumetric flow rate and cross-sectional area, based on the principle of conservation of mass for incompressible flow. (Pritchard & Mitchell, 2016)

The volumetric flow rate is calculated as: $Q = \frac{V}{t}$

The velocity of flow at any section is given by:

$$V = \frac{Q}{A}$$

For an incompressible fluid, continuity must be satisfied:

$$A_1 V_1 = A_2 V_2$$

1.3.3 Test procedure

- ✚ The experimental setup was arranged following the lab guidelines, making sure that all the pipe connections were tight and leak-free.
- ✚ The flow system was activated, allowing water to move through the apparatus until a steady flow was reached.
- ✚ The control valve was then modified to establish a specific flow rate in the system.
- ✚ A specific amount of water was gathered in a measuring tank over a timed duration using a stopwatch.
- ✚ This procedure was repeated multiple times under the same flow conditions to enhance accuracy, and an average time was determined.



Fig: Venturi meter



Fig: Piezometer Tube

1.4 Results and Discussion

1.4.1. Table of raw data

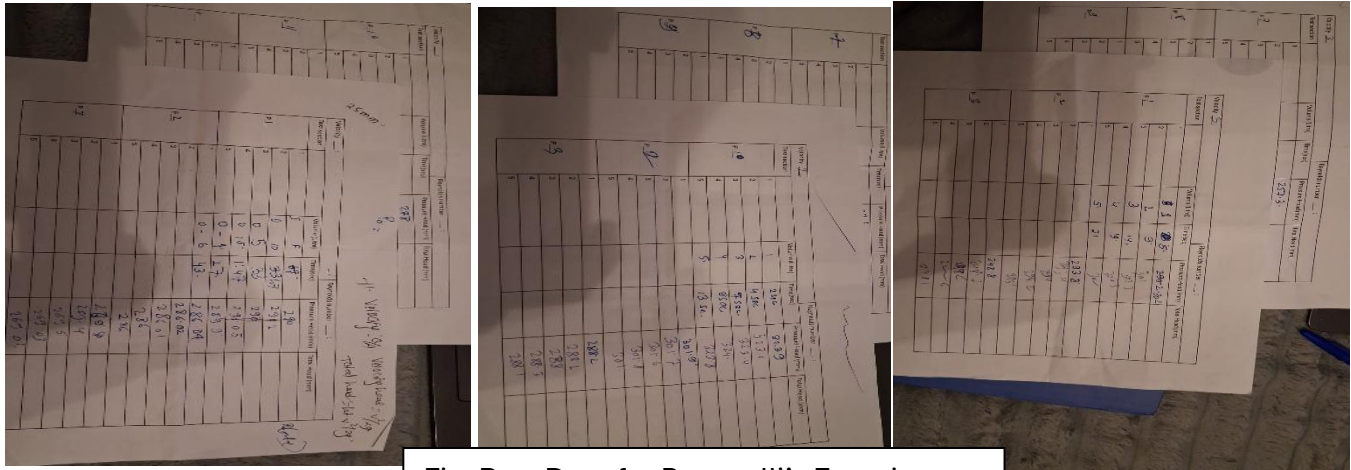


Fig: Raw Data for Bernoulli's Experiment

1.4.2. Table of Results

for velocity one									
test section	volume	Dia	area	time	pressure head	flow rate	velocity	velocity head	total head
P1	0.008	0.03	0.00053	55.4	0.29	0.0001444	0.272122	0.003774	0.2937742
P2		0.02	0.00039		0.285		0.369914	0.006974	0.2887742
P3		0.02	0.00022		0.29		0.644057	0.021142	0.2937742
P4		0.01	0.00015		0.294		0.938516	0.044894	0.2977742
P5		0.02	0.00018		0.283		0.796178	0.032309	0.2867742
P6		0.02	0.00024		0.295		0.60065	0.018388	0.2987742
P7		0.02	0.00031		0.289		0.46921	0.011221	0.2927742
P8		0.02	0.00039		0.274		0.373243	0.0071	0.2777742
P9		0.02	0.00047		0.276		0.306454	0.004787	0.2797742
P10		0.03	0.00053		0.278		0.272114	0.003774	0.2817742
P11		0.03	0.00053		0.294		0.272114	0.003774	0.2977742
Summary									
Average head		0.289956051							
Minimum Total head		0.286774233							
Maximum total head		0.297774233							
% Variation		3.793678368							
Interpretation		< 10% = good, 10-20% = fair							
		> 20% = poor							
Assessment		Good							

for velocity 2									
test section	volume	Dia	area	time	pressure head	flow rate	Velocity	velocity head	total head
P1	0.003	0.03	0.00053	13.6	0.299	0.00022059	0.415687	0.008807	0.3078071
P2		0.02	0.00039		0.293		0.565075	0.016275	0.3018071
P3		0.02	0.00022		0.294		0.983881	0.049339	0.3028071
P4		0.01	0.00015		0.296		1.433706	0.104766	0.3048071
P5		0.02	0.00018		0.293		1.216267	0.075398	0.3018071
P6		0.02	0.00024		0.299		0.917572	0.042912	0.3078071
P7		0.02	0.00031		0.295		0.71678	0.026186	0.3038071
P8		0.02	0.00039		0.296		0.570178	0.01657	0.3048071
P9		0.02	0.00047		0.298		0.468149	0.01117	0.3068071
P10		0.03	0.00053		0.299		0.41569	0.008807	0.3078071
P11		0.03	0.00053		0.298		0.41569	0.008807	0.3068071

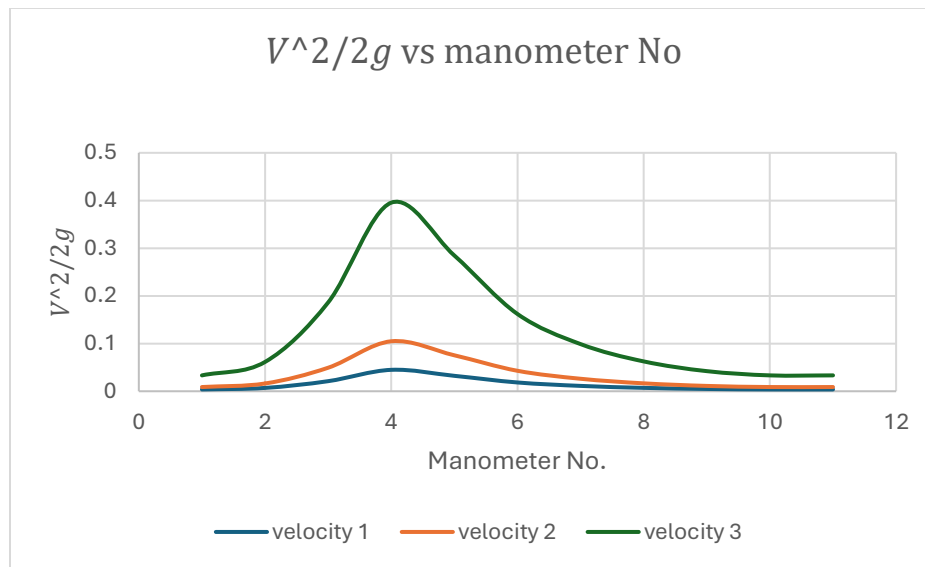
Summary	
Average head	0.305170738
Minimum Total head	0.301807101
Maximum total head	0.307807101
% Variation	1.966112495
Interpretation	< 10% = good, 10-20% = fair > 20% = poor
Assesment	good

Velocity 3									
test section	volume	dia	area	time	pressure head	flow rate	velocity	velocity head	total head
P1	0.003	0.03	0.00053	7	0.323	0.00042857	0.80762	0.033244	0.3618474
P2		0.02	0.00039		0.301		1.097848	0.061431	0.3398474
P3		0.02	0.00022		0.288		1.911519	0.186234	0.3268474
P4		0.01	0.00015		0.29		2.785454	0.395451	0.3288474
P5		0.02	0.00018		0.289		2.363007	0.284597	0.3278474
P6		0.02	0.00024		0.279		1.782691	0.161977	0.3178474
P7		0.02	0.00031		0.28		1.392585	0.098843	0.3188474
P8		0.02	0.00039		0.282		1.107761	0.062545	0.3208474
P9		0.02	0.00047		0.26		0.909536	0.042164	0.2988474
P10		0.03	0.00053		0.272		0.807617	0.033244	0.3108474
P11		0.03	0.00053		0.273		0.807617	0.033244	0.3118474

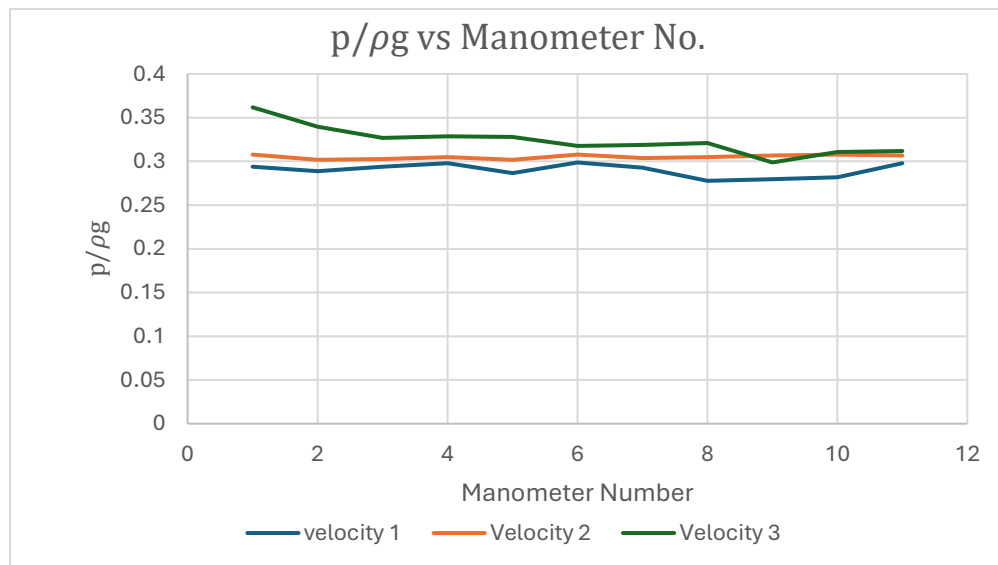
Summary	
Average head	0.324029168
Minimum Total head	0.29884735
Maximum total head	0.31184735
% Variation	4.011984499
Interpretation	< 10% = good, 10-20% = fair > 20% = poor
Assesment	good

All the Units are in meter

1.4.3 Graphs



Graph 1: graph between $V^2/2g$ vs manometer number



Graph 2: graph between $P/\rho g$ with and manometer No

1.5. conclusion and discussion of any discrepancies observed.

1.5.1 Conclusion

The validity of the Bernoulli equation was evaluated by checking if the total head (which is the sum of pressure head and velocity head) stayed constant across the test sections for various flow speeds. (Elger et al., 2020)

For Velocity 1, the total head was between about 0.287 m and 0.298 m, with an average of 0.291 m and a percentage variation of 3.79%. This small change shows a strong alignment with Bernoulli's equation, indicating that the total mechanical energy in the tube stays almost constant. For Velocity 2, the total head fluctuated from around 0.303 m to 0.309 m. Even though there are minor changes compared to Velocity 1, the variation is still low, which reflects good alignment with what we expect theoretically. For Velocity 3, the total head was between approximately 0.299 m and 0.362 m. This larger variation implies that the differences from the ideal Bernoulli behavior become more apparent at higher speeds, probably because of increased friction losses and uncertainties in the experiments. In summary, the findings show that Bernoulli's principle is generally well upheld, with better alignment seen at lower speeds.

1.5.2 Discrepancies Observed

Some differences from Bernoulli's equation were noticed, particularly at greater speeds. These differences are mostly caused by friction losses in the pipe, which lower the total head. When the flow rates are higher, turbulence increases energy losses, leading to bigger variations. (Pritchard & Mitchell, 2016)

Moreover, errors in measurement (like reading the piezometer levels or timing mistakes) and potential air bubbles in the tubes might have influenced the pressure readings. Small losses from fittings and variations in pipe area also play a role in the inconsistencies. (Elger et al., 2020)

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Part 2- Hydraulic Modelling of Flow Over a Weir in an Open Channel

2.1. Objective

To simulate and study the flow over a weir in an open channel so we can find out the head-discharge relationship and assess the discharge coefficient. (Te Chow 1959)

2.2. Apparatus used

1. point gauge
2. Hydraulic flume with a broad-crested weir
3. Stopwatch
4. Scale

2.3. Theory

A broad-crested weir is utilized in open channel flow to gauge discharge. Under perfect conditions, the flow over the weir crest attains critical depth, which enables the discharge to be connected to the upstream head. (Henderson 1966)

Assuming negligible head loss between upstream and the weir crest:

$$H_{weir} = H_{upstream}$$

where total head is $H = z + \frac{v^2}{2g}$

At the crest, flow is critical:

$$h_c = \left(\frac{q^2}{g}\right)^{1/3}, q = \frac{Q}{b}$$

Taking the crest as datum:

$$\frac{3}{2} \left(\frac{q^2}{g}\right)^{1/3} = h_0 + \frac{q^2}{2gh_1^2}$$

Rearranging for discharge per unit width:

$$q = \left(\frac{2}{3}\right)^{3/2} \sqrt{g} \left(h_0 + \frac{q^2}{2gh_1^2}\right)^{3/2}$$

Total discharge =

$$Q_{ideal} = 1.705 b \left(h_0 + \frac{Q_{ideal}^2}{19.62 b^2 h_1^2}\right)^{3/2}$$

Accounting for real flow:

$$C_d = \frac{Q}{Q_{ideal}}$$

2.3.1 Test Procedure

- ✚ The open channel flow flume was checked to make sure it was level and clear of any debris. The broad-crested weir was verified to be securely placed in the test section.
- ✚ The pump was turned on, and the flow control valve was opened slowly to let water into the channel. We were careful to prevent any sudden surges.
- ✚ The downstream tailgate was adjusted until we could see a steady water surface profile upstream of the weir.
- ✚ We let the flow stabilize for a few minutes. We considered the conditions steady when the water level showed very little fluctuation.
- ✚ The point gauge was first set to zero at the level of the weir crest. Then, we carefully measured the upstream depth h_1 and the head over the weir h_0 , making sure the probe just touched the water's surface.
- ✚ Discharge was measured by redirecting flow into a volumetric tank. We recorded the time it took to collect a known volume using a stopwatch.
- ✚ We took at least two readings of volume and time for each flow condition and averaged them to enhance accuracy.
- ✚ The flow rate was then adjusted using the control valve, and we repeated the above measurements for several different flow conditions.
- ✚ Throughout the experiment, we were careful to avoid disturbing the flow while taking readings.
- ✚ After finishing all the measurements, we gradually closed the valve, turned off the pump, and drained the flume.

2.4 Result and Discussion

2.4.1. Table of Raw Data

trial no.	weir height	weir board	upstream water depth	water depth above the weir	volume collected	time
1)	10 cm	10 cm	11.6 mm	10 mm, 3 mm, 10 mm	5 L	17 sec
2)	10 cm	12.5 cm	12.3 mm	14 mm	15 L	28 sec

Interval width of Channel = 76.1 mm
 $L = 35 \text{ cm}, B = 7.5 \text{ cm}, h = 110 \text{ mm}$
 $h = 76.87 \text{ mm}$
 $h = 76.37 \text{ mm}$

Fig: raw data for open channel flow

2.4.2 Calculation

trial no	Volume	time	upstream water depth (h1)	Upstream water depth above weir (h0)	head	Discharge	Theoretical Discharge	coefficient of discharge	% Error	$h^{3/2}$
1	0.005	4	0.116	0.1	0.016	0.0013	0.003043	0.75624	58.922	0.002
2	0.015	9	0.123	0.098	0.025	0.0038	0.003923	0.9559	4.4099	0.004

All units are in meter

2.4.3 Site photos



2.4.4 Graph

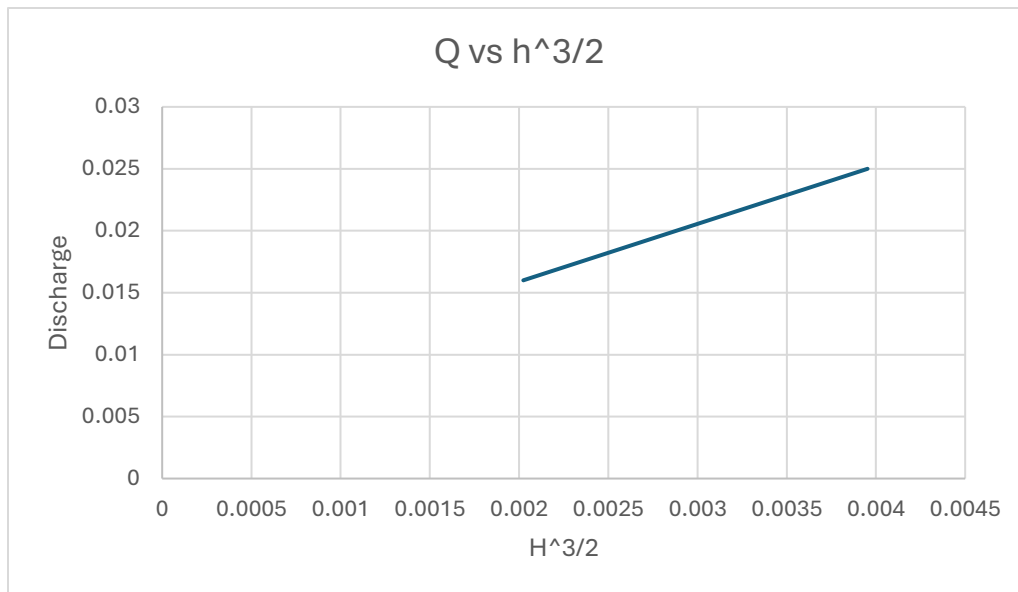


Fig: graph between discharge vs $h^{3/2}$

2.4.5 Comparison of Experimental Results with Theory

The experimental discharge coefficient, C_d , for the broad-crested weir was compared to theoretical and literature values. The percentage error was determined using:

$$\% \text{ error} = \left| \frac{C_{d,\text{exp}} - C_{d,\text{theory}}}{C_{d,\text{theory}}} \right| \times 100$$

For a broad-crested weir, the discharge coefficient is influenced by several factors including the upstream head, the length of the crest, the shape of the crest, the roughness, and the flow conditions. Research indicates that broad-crested weirs are frequently utilized as structures for measuring discharge, and their discharge coefficient is not fixed; it changes based on the geometry and hydraulic conditions. (Tracy 1957)

The U.S. Bureau of Reclamation mentions that to measure weir discharge, you need to check the upstream head in relation to the crest and apply the right equations or tables for the specific type of weir. (Ruff 2008)

2.4.6 Comparison with Released Cd Values

The coefficients of discharge (C_d) for the broad-crested weir, which were found through experiments, were compared to the values that are published in standard hydraulic literature. Usually, for broad-crested weirs, the coefficient of discharge falls between 0.95 and 1.00 when the flow conditions are ideal. (Chadwick et al. 2021; Te Chow 1959)

In this experiment, the values of C_d recorded were 0.756 for Trial 1 and 0.956 for Trial 2. The second trial closely matches the published value, showing a percentage error of just 4.41%, which suggests that the experimental conditions were reliable. On the other hand, the first trial has a much lower C_d value and a high percentage error of 58.92%, indicating significant experimental errors. The discrepancies seen in Trial 1 could be due to mistakes in measuring head, unstable flow conditions, or the flow not reaching critical conditions over the crest. In contrast, Trial 2 seems to meet the theoretical flow assumptions better, leading to a discharge coefficient that is much nearer to the expected range. In summary, the results show that while experimental values can closely match published data, the accuracy heavily relies on correct measurement methods and stable flow conditions.

2.4.7 Observation and sources of error

During the experiment, it was noted that the discharge went up as the head over the weir rose. This matches what is expected from the theory of broad-crested weirs, where the discharge is heavily influenced by the upstream head.

The differences in the calculated discharge coefficient can be linked to various sources of experimental error. These errors include inaccurate head measurements caused by parallax error and fluctuations in the water surface near the measuring point, which can result in incorrect readings of the upstream head. Additionally, if the measuring gauge is not positioned correctly, it may have caused errors in the recorded values. Friction losses along the broad crest and the roughness of the weir's surface can also lower the actual discharge compared to what is predicted theoretically. Furthermore, non-uniform approach velocity

and the chance that the flow didn't completely reach critical conditions at the crest add to the discrepancies from ideal assumptions. Errors in timing and volume measurements during the discharge calculation, leakage around the weir plate, and unsteady flow conditions during readings might have also impacted the accuracy of the results. All these factors together result in differences between the experimental and theoretical values of the discharge coefficient.(Chadwick et al. 2021; Pritchard and Mitchell 2016)

2.4.8 Dimensional and Similarity Considerations

According to the experimental findings, the discharge over the broad-crested weir rose as the head increased. In Trial 1, the head measured 0.016 m with a discharge of 0.00125 m³/s, whereas in Trial 2, the head increased to 0.025 m and the discharge rose to 0.00375 m³/s. This observation corroborates the theoretical relationship for broad-crested weirs, indicating that discharge is proportional to $H^{3/2}$.(Chadwick et al. 2021; Te Chow 1959).The computed $H^{3/2}$ values rose from 0.00202 to 0.00395, indicating that the flow adheres to the anticipated dimensional relationship. Given that flow over a weir is primarily governed by gravity, the Froude number serves as the most crucial similarity parameter. At the crest, the flow nears critical conditions, where $Fr \approx 1$. Although the effects of the Reynolds number are not as pronounced, minor variations may arise due to viscosity, surface roughness, and errors in experimental measurements.(Chadwick et al. 2021)

2.5 Conclusion

The experiment effectively explored the flow over a broad-crested weir and determined the connection between head and discharge. The findings indicated that discharge rose as head increased, aligning with the theoretical relationship $Q \propto H^{3/2}$. In the second trial, the coefficient of discharge was in good agreement with published values, whereas the first trial showed discrepancies due to experimental errors and non-ideal flow conditions. In summary, the experiment validated the theoretical assumptions for broad-crested weirs and emphasized the need for precise measurements and stable flow conditions to achieve reliable results.(Chadwick et al. 2021)

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Part 3. Centrifugal Pump Performance & Selection

3.1 Engineering Scenario

Flow rate, $Q = 25 \text{ L/s} = 0.025 \text{ m}^3/\text{s}$

Static head, $H_s = 18 \text{ m}$

Pipe length, $L = 120 \text{ m}$

Pipe diameter, $D = 0.15 \text{ m}$

Friction factor, $f = 0.02$

Water density, $\rho = 1000 \text{ kg/m}^3$

Gravity, $g = 9.81 \text{ m/s}^2$

3.2 Pump Selection and criteria

3.2.1 Identifying which meets the required flow and head

The required flow rate is:

$$Q = 25 \text{ L/s} = 0.025 \text{ m}^3/\text{s}$$

The pipe area is:

$$A = \frac{\pi D^2}{4} = \frac{\pi (0.15)^2}{4} = 0.0177 \text{ m}^2$$

The flow velocity is:

$$V = \frac{Q}{A} = \frac{0.025}{0.0177} = 1.41 \text{ m/s}$$

The friction head loss is:

$$\begin{aligned} H_f &= f \frac{L}{D} \frac{V^2}{2g} \\ H_f &= 0.02 \times \frac{120}{0.15} \times \frac{1.41^2}{2(9.81)} \\ H_f &= 1.63 \text{ m} \end{aligned}$$

Therefore, the total dynamic head is:

$$H_{system} = H_{static} + H_{friction}$$

$$H_{system} = 18 + 1.63 = 19.63 \text{ m}$$

The pump is required to provide a flow rate of $0.025 \text{ m}^3/\text{s}$ at a head of approximately 19.63 m. Among the options, Pump B is the most appropriate choice as its Best Efficiency Point (BEP) is 27 L/s, which is nearest to the desired flow rate of 25 L/s, and it also exhibits the highest efficiency.

3.2.2 Determining the operating efficiency and power requirement

For Pump B:

$$\eta = 82\% = 0.82$$

The power requirement is calculated using:

$$P = \frac{\rho g Q H}{\eta}$$

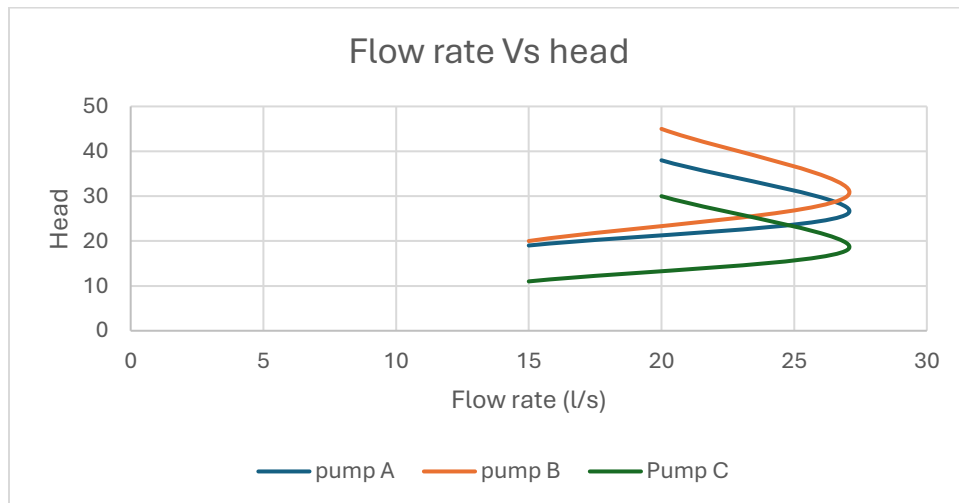
$$P = \frac{1000 \times 9.81 \times 0.025 \times 19.63}{0.82}$$

$$P = 5869 \text{ W}$$

$$P = 5.87 \text{ kW}$$

Therefore, the required pump power is approximately **5.87 kW**.

3.2.3 Graphs



3.3 Discussion

3.3.1 Proximity to BEP

Pump B is chosen since it runs nearer to its Best Efficiency Point (BEP), which is 27 L/s, while the needed flow rate is 25 L/s. Working near the BEP guarantees better efficiency, less vibration, and greater reliability. On the other hand, Pump A and Pump C would function farther from their BEP, raising the chances of poor performance.

3.3.2 Risk of operating too far left/right of curve & Energy consumption implications

Running a pump too far to the right on its curve can lead to lower efficiency and might result in more vibration, noise, overheating, and wear on the machine. This can also make the pump less reliable over time. Because the system runs all the time, how much energy it uses is important. Pump B is the most efficient at 82%, meaning it will use less power and cost less to operate than the other pump choices. (Zhu et al. 2021)

Running a centrifugal pump too far from its Best Efficiency Point (BEP) can really impact how well it works and how much energy it uses. If the pump is working to the left of the curve (low flow), it might face issues like internal recirculation, more vibration, and overheating because of unstable flow conditions. On the other hand, if it's to the right of the curve (high flow), it can create too much turbulence, lower head, and flow separation, which all lead to less efficiency. In both situations, the pump needs more energy to provide the necessary output, which means higher operating costs, especially in continuous systems. If the pump runs away from the BEP for a long time, it can also cause more mechanical wear and shorten the pump's lifespan. So, it's important to choose a pump that works close to its BEP to keep energy efficiency. (Al-Obaidi and Shaban 2024a)

3.3.3 Cavitation considerations

Cavitation is another important factor to think about. When the suction pressure drops too low, vapor bubbles can develop inside the pump and then collapse near the impeller. This can lead to noise, vibrations, decreased performance, and even potential damage. To lower the chances of cavitation, it's best to keep the suction pipe short and straight, minimize suction losses, and ensure there is enough net positive suction head available. (Al Mamun et al. 2018)

3.4 Conclusion and observation

The examination of the pumping system indicates that the necessary operating conditions include a flow rate of $0.025 \text{ m}^3/\text{s}$ and a total dynamic head of about 19.63 m. Out of the options available, Pump B stands out as the best choice since its Best Efficiency Point (BEP) of 27 L/s is the nearest to the required flow rate of 25 L/s, which means it can work under nearly optimal conditions.

Running a pump too far from its Best Efficiency Point (BEP) can really impact how well it works and how much energy it uses. If the pump is working to the left of the curve (which means low flow), it might cause problems like internal recirculation, more vibration, and overheating because the flow isn't stable. On the other hand, if it's to the right of the curve (high flow), there can be too much turbulence, less head, and flow separation, which can make it less efficient. In both situations, the pump needs more energy to keep producing the same output, which can lead to higher operating costs, especially in systems that run continuously. If a pump runs away from the BEP for a long time, it can also cause more wear and tear on the mechanical parts, shortening the pump's lifespan.(Al-Obaidi and Shaban 2024b)

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